

Self-Attention Mechanisms in Transformer Architectures

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Abstract:

Transformer networks rely heavily on scaled dot-product attention to compute token associations. Given Query (Q), Key (K), and Value (V) matrices, attention weights are derived via $\text{Softmax}(QK^T / \sqrt{d_k})$. While standard full-attention incurs quadratic computational complexity $O(N^2)$ relative to sequence length, recent breakthroughs like FlashAttention use fused GPU memory kernels and IO-awareness to overcome memory-bandwidth bottlenecks without numerical approximation.

Key Research Contributions:

- Theoretical analysis of multi-head attention versus single-head representation capacity.
- Rotary Positional Embeddings (RoPE) for relative position awareness across long context windows.
- Empirical evaluation on perplexity benchmarks across billion-token open-source pretraining datasets.